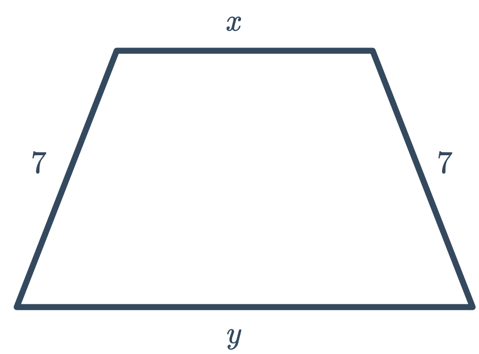
Practical problems with equations

1 When a number is added to both the numerator and denominator of $\frac{1}{5}$, the result is $\frac{3}{7}$. Find the number.

2 The perimeter of the given trapezoid is 27 cm.

a Write an equation in terms of x and y for the perimeter.

b Find the value of y, if x is 5 cm.



3 Valentina tries to guess how many people are at a concert, but she guesses 400 too many. Kenneth guesses 150 too few. The average of their guesses is 3625. Find the exact number of people at the concert.

4 4 divided by the sum of x and 6 equals the quotient of 2 and the difference of x and 6. Find the value of x.

5 Sharon, a tennis player has won 54 out of 78 matches in her career. Find the number of matches she must win in a row to raise her win percentage to 75%.

6 It takes Isabelle 16 hours to build a playhouse. If she and Harry work together, they can finish building a playhouse in 12 hours. Find the number of hours that it will take Harry to build a playhouse by himself.

7 A receptionist can type documents 5 times as fast as his assistant. Working together, they can type up a day's worth of documents in 6 hours. On a day that the assistant is absent from work. Find the number of hours that it will take the receptionist to type up the day's documents on his own.

8 The sum of the reciprocals of two integers separated by 7 is equal to 11 times the reciprocal of their product. Find:

a The smaller integer

b The larger integer

9 Eileen's bathtub can be filled in 6 minutes and can subsequently be drained in 15 minutes when it is unplugged. Find the number of minutes that it will it take to fill the bathtub if it is left unplugged while being filled.

10 Gwen can paint the average house in 12 hours, while Maria can do it in 6 hours. Find the number of hours that it would take to paint a house if they both worked together.

11 Sisters Judy and Tara specialise in two different triathlon events. Judy finds that her average cycling speed is 8 mph faster than Tara's average running speed. Judy can cycle 22 miles in the same time that it takes Tara to run 11 miles. Find:

a The running speed of Tara

b Judy's average cycling speed.

12 The current of a river is 3 mph. Salmon can swim 30 miles downstream (with the current) in the same amount of time it takes for it to swim 15 miles upstream (against the current). Salmon normally swim at $n \text{ mph}$ with no current.

a Complete the following table:

	Distance (miles)	Rate	Time
Upstream	15	$n-3$	
Downstream	30	$n+3$	

b Find the value of n.

13 A river is flowing at a speed of 9 mph. Salmon can swim 25 miles downstream (with the flow) in the same amount of time it takes for it to swim 20 miles upstream (against the flow). If salmon normally swim at $n \text{ mph}$ with no current, solve for n.

14 When a new car was released on the market, car journalists Valerie and Victoria were invited to take the car for a test drive. Each took their car into the mountains, where Valerie covered a distance of 22 miles in the same time that Victoria drove 15 miles. Valerie drove 21 mph faster than Victoria.
Find the speed of:

a Victoria

b Valerie

15 A plane made a return journey, where the speed of the wind was a constant n mph going and coming back. On its first leg, it traveled 700 miles with the wind, and took the same number of hours to fly back 600 miles against the wind. If there were no wind, the plane would have traveled 299 mph. Find the speed of the wind.

16 A plane averaged 380 mph on a trip going east from Capital City to Springfield, but only 350 mph on the return trip from Springfield back to Capital City. The total flying time in both directions was $t = 6.5$ hrs.

a Let k be the proportion of the total flight time t, so that the flight from Capital City to Springfield takes kt hours, where $0 < k < 1$. Write an expression for the duration of the return trip from Springfield to Capital City in terms of k and t.

b In terms of k and t, write an equation for the following:

i One-way distance d from Capital City to Springfield

ii Distance d of the return trip from Springfield to Capital City

c Find the value of k.

d Find the one-way distance between Capital City and Springfield rounded to the nearest whole mile.

17 Solve for x in the following statements:

a The sum of 8 and $12x$ is equal to 92.

b The product of 5 and the sum of x and 7 equals 50.

c The 5th multiple of the difference of 5 from x is equal to -15.

18 John and Uther do some fundraising for their sporting team. Together they raised $\$403$. If John raised $\$m$, and Uther raised $\$71$:

a Write an equation that represents the relationship between the amounts each contributed.

b Find the value of m.

19 Kate and Isabelle do some fundraising for their sporting team. Together they raised $\$600$. If Kate raised $\$272$ more than Isabelle, and Isabelle raised $\$p$:

a Write an equation in terms of p that represents the relationship between the different amounts, and then solve for p.

b How much did Kate raise?

20 Let x be the smallest of three consecutive even integers. Write an expression for:

- **a** The second integer
- **b** The third integer
- **c** The sum of the first and third consecutive even integers

21 Find three consecutive even integers whose sum is 36.

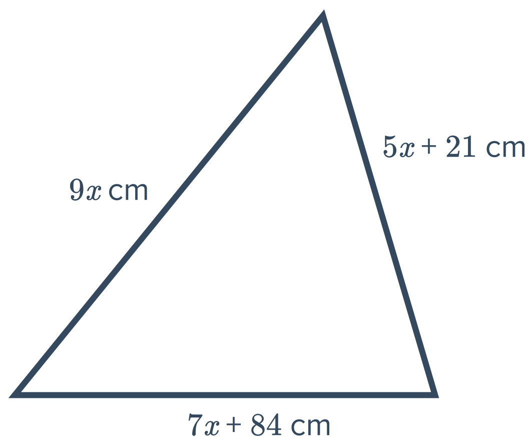
22 Find four consecutive odd integers whose sum is 64.

23 One quarter of a number is equal to triple that number less 22. Let the unknown number be x.

- **a** Write an equation in terms of x.
- **b** Find the value of x.

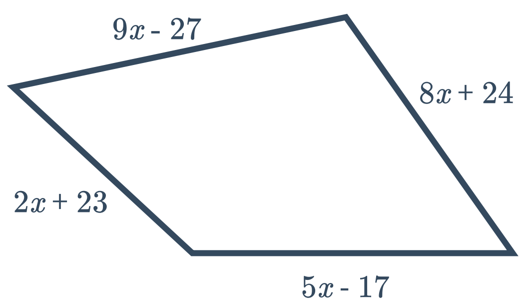
24 Consider the following triangle with a perimenter of189 cm:

- **a** Write an equation in terms of x.
- **b** Find the value of x.



25 Consider the following quadrilateral with a perimeter of 315 cm:

- **a** Write an expression for the perimeter in terms of x.
- **b** Find the value of x.



26 Vanessa is cutting out a rectangular board to construct a bookshelf. The board is to have a perimeter of 48 inches, and its length is to be 3 inches shorter than double the width. Let x be the width of the board.

a Write the expression for the length in terms of x.

b Find x.

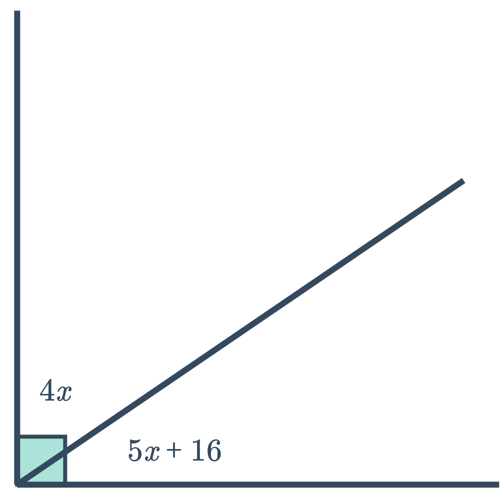
c Hence, find the length of the board.

27 Three consecutive integers are such that the sum of the first and twice the second is 12 more than twice the third. Find the three integers.

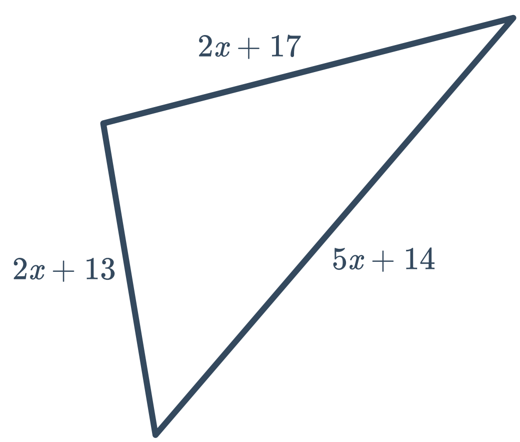
28 Consider the angles marked in the following diagram:

a Find the value of x.

b Find the measure of the angle marked by $4x$.



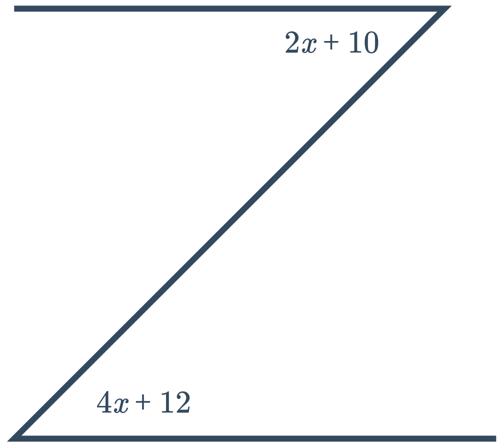
29 Given that the perimeter of this triangle is 98 cm, find x.



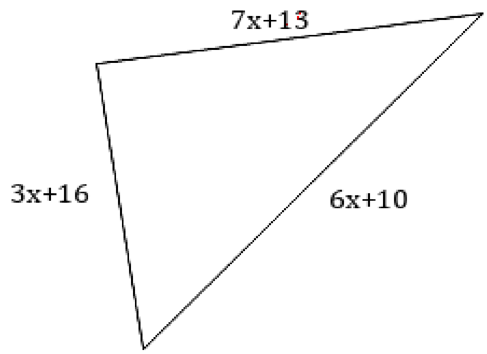
30 Consider the two angles given in the diagram:

a Find the value of x.

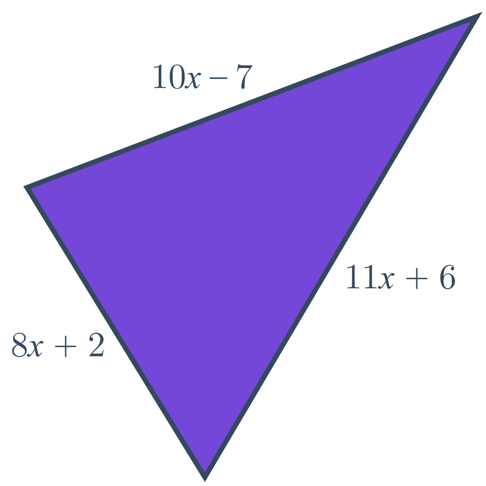
b Hence, find the size of one of the angles.



31 Given that the perimeter of the triangle is 231 cm, find x.



32 Given that the perimeter of the triangle is 146 cm, find x.



33 The cost of a cricket ball is 136 cents more than the cost of a rugby ball. Let $\$x$ be the cost of a cricket ball and $\$y$ be the cost of a rugby ball.

a Express x in terms of y

b If 8 cricket balls and 9 rugby balls cost $\$119$, write an equation in terms of x and y.

c Find the value of y.

d Find the value of x.

34 One side of a parallelogram is 4 cm shorter than an adjacent side. Let x cm be the length of the shorter side and y cm be the length of the adjacent side to x. The perimeter of this parallelogram is 12 cm.

a Express x in terms of y.

b Write an equation in terms of y.

c Find the value of y.

d Hence, find the value of x.

35 Beth is 7 times as old as James. Let x be the present age of Beth in years and y be the present age of James in years.

a Express x in terms of y.

b In 2 years time, Beth will be 5 times as old as James. Write an equation in terms of x and y.

c Find the value of y.

d Hence, find the value of x.

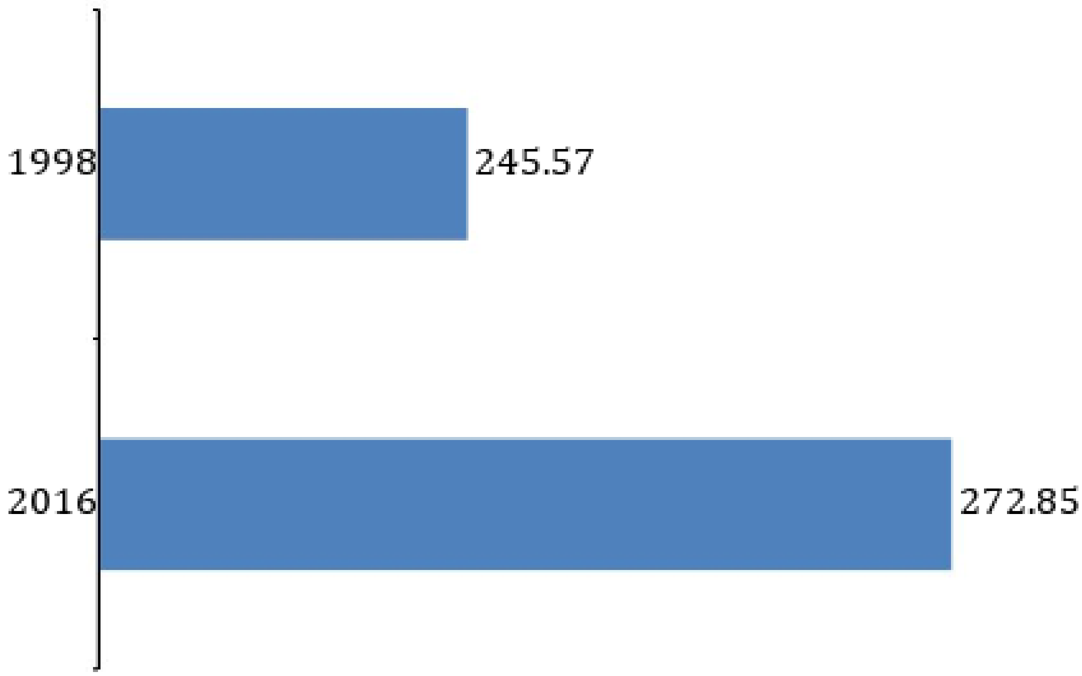
36 The relationship between F degrees Fahrenheit and C degrees Celsius is $F = 1.8C + 32$. Find the temperature in degrees Celsius that is equivalent to 51.8 degrees Fahrenheit.

37 Find the unknown number in the following statements:

a If 11 is subtracted from a number the result is -7.

b The quotient of a number and -3 is -20.

38 The following bar graph shows the average weekly earning difference, in dollars, between men and women working full-time.
The data in the bar graph can be described by the mathematical model $D = 1.84n + 238$, where D represents the average weekly difference in earnings n years after 1998.



a Use the formula to estimate the average weekly earning difference, D, in 2016.

b State whether the formula underestimate or overestimate the average weekly earning difference in 2016.

c The formula underestimated the average weekly pay gap by how much?

d If the trend described by the formula continues, how many years after 1998 will the average weekly earning difference be $\$296.88$?

e Hence, if the trends continue in the way modeled by the formula, in what year would the average weekly earning difference between men and women will be $\$296.88$?

39 A Payroll Officer has been told to distribute a bonus to the employees of a company worth 14% of the company's net income. Since the bonus is an expense to the company, it must be subtracted from the income to determine the net income. If the company has an income of $\$120\,000$ before the bonus, then the Payroll Officer must solve the following equation to find the bonus B:

$$B = 0.14\,(120\,000 - B)$$

Find B, correct to two decimal places.

40 Currently, Patricia's father is 56 years older than Patricia. 3 years ago, her father was 5 times

41 A commercial airplane has a total mass at take off of $51\,000$ kg. The luggage and fuel are $\frac{1}{3}$ the mass of the unloaded plane, and the crew and passengers are $\frac{1}{4}$ the mass of the fuel and luggage. Find the mass of the unloaded plane.

42 A website advertises properties for sale. If the property is sold within the month, the website charges $\$130$ for the advertisement, but if the property is not sold within the month, the website pays the advertiser $\$20$. In a particular month, 58 advertisements were placed, and a total of $\$3040$ was made across all advertising. Find the number of advertisements that resulted in a sale of the property.

43 Marge is looking at accommodation prices in Paris. One particular hotel charges $\$184.70$ for the first night, and then $\$153.97$ for every additional night. Marge has a budget of $\$1108.52$. Find the number of nights she can afford to stay.

44 To save water, a household decides to install one large and one smaller water tank. The smaller tank will hold 160 L less than the larger tank, and when the larger tank is $\frac{3}{5}$ full, it will be able to completely fill the smaller tank. Find the capacity of the smaller tank.

45 Sophia is the youngest child in her family. She has three older brothers of ages 19, 16 and 12, and a sister who is eight years older than Sophia. The average age of the five children is 15 years. Find Sophia's age.

46 A two-digit number is such that the tens digit is 7 more than the ones digit, and the number itself is 9 times the sum of its digits.

a Find the tens digit.

b Hence, find the number.

47 Edward and Roald both leave Sydney at the same time to go on holidays. Edward travels by car and drives at an average speed of 80 km/h. Roald is going 3080 km further than Edward, so he travels on a plane which moves at an average speed of 850 km/h. Find the number of hours each take for their journey.

48 A rectangular rooftop has an area of 260 square feet. The length of the rooftop is 13 feet more than twice the width, w, of the rooftop. Write an equation in terms of w that models the situation.

Additional questions

49 Solve the following equations:

a $4x = 8$ b $5m = -15$ c $3p = 12$ d $2k = -18$

50 Solve the following equations:

a $8m + 9 = 65$

b $-10 + 3k = 5$

c $7 - 8t = 15$

d $-2d + 10 = 24$

e $3f - 8 = f$

f $10r + 4 = 6r$

g $4w + 24 = w + 15$

h $-72 - 9p = -32 - p$

51 Solve the following equations:

a $\dfrac{x}{2} - 3 = 1$

b $\dfrac{3x - 15}{5} = 9$

c $\dfrac{3x}{5} - \dfrac{4x}{6} = -5$

d $\dfrac{10 + x}{10} = \dfrac{x + 4}{10}$

52 Solve the following equations:

a $-2\left(-3h - 5\right) = 34$

b $9x + 6x + 7 - 3x = 55$

c $4x + 2 + 9x - 12 + 6x + 7 = 130$

d $m + m + 3 + 12 = 13 + m + m + m$

e $x + x + x + x + 2 + 5 = x + 4 + x + x$

f $p + p - 3 + 5 = p - p - p + 1 - 5$

g $-k - k - k - 6 = -k - k - k - k + 6$

h $4\left(x - 15\right) = x$

i $5\left(x - 16\right) = x - 16$

j $3\left(x - 3\right) = 2\left(x + 1\right)$

k $2\left(3x - 4\right) = 2\left(x + 4\right)$

l $3y = -13 - 4\left(2y + 5\right)$

m $x - 6 - 7\left(x + 3\right) = 21 - 9\left(x - 18\right)$

53 Solve the following equations:

a $3\left(x + \dfrac{3}{2}\right) + 3 = -\dfrac{15}{2}$

b $x + \dfrac{3x + 4}{2} = 3$

c $\dfrac{5x - 1}{3} - \dfrac{2x - 4}{5} = -1$

d $10x - \dfrac{33}{2} = 4x$

e $\dfrac{14k - 76}{2} - 2 = 2k$

f $\dfrac{3y + 3}{3} - \dfrac{2y - 4}{4} = 2y - 10$

g $5x - 2$

54 Determine the number of solutions of the following equations:

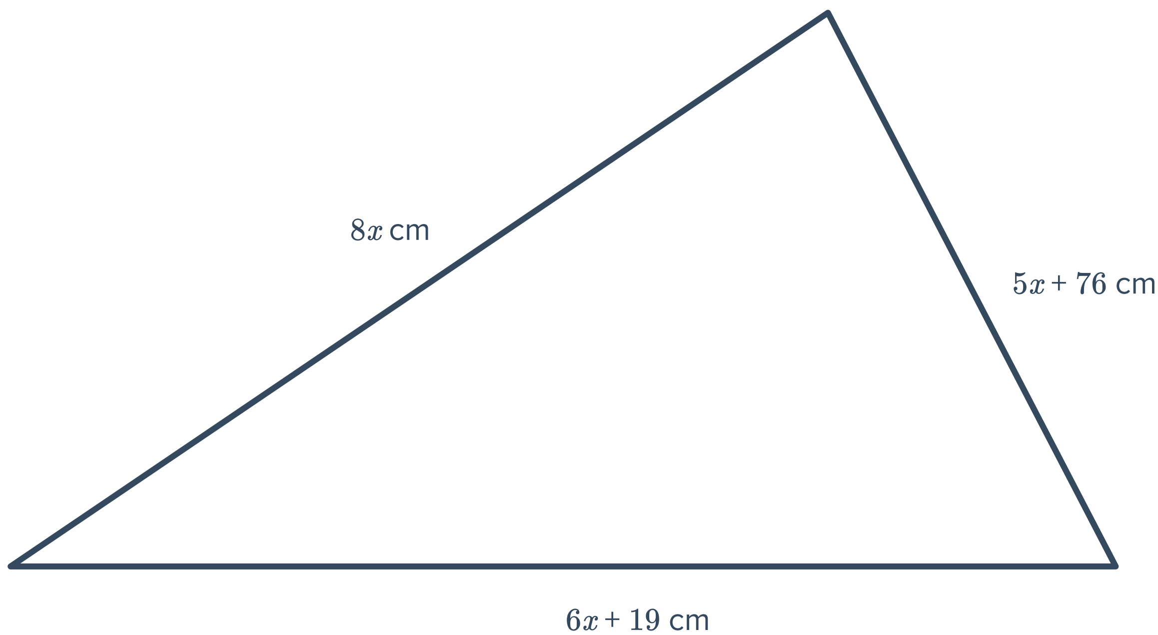
a $7(7+x)=2x+28$

b $10(5+x)=10(x+5)$

c $-31-3(x-9)=2(x-2)-5x$

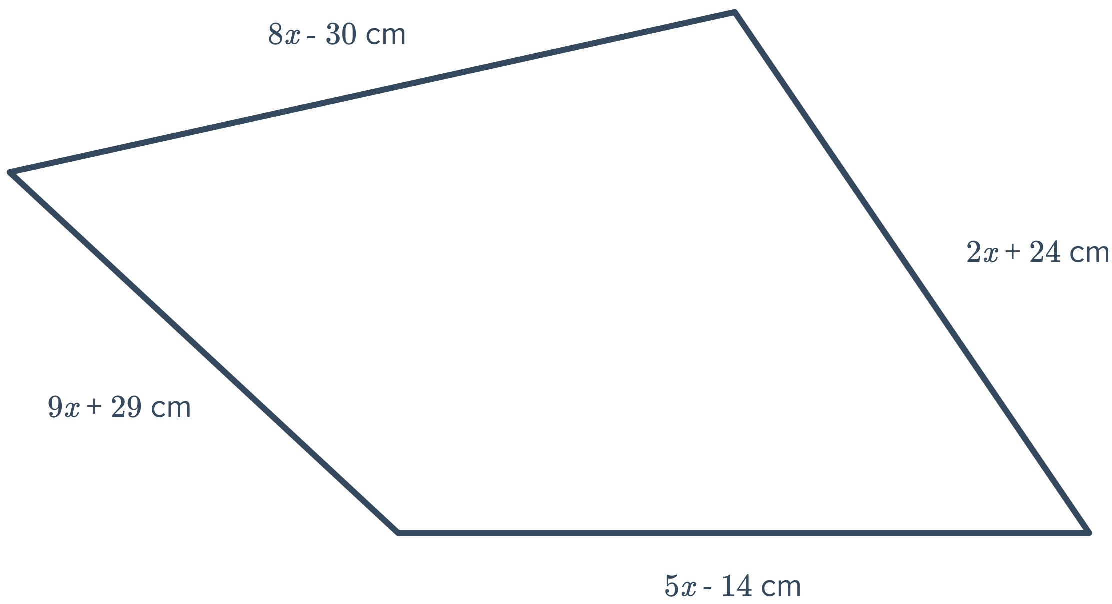
55 The product of 6 and the sum of x and 4 is equal to 42. Find the value of x.

56 Consider the given triangle which has a perimeter of 171 cm.



Solve for the value of x.

57 Consider the given quadrilateral which has a perimeter of 393 cm.



Solve for the value of x.

58 Ursula is constructing a tabletop from a rectangular piece of board. The tabletop is to have a perimeter of 68 inches, and its length is to be 5 inches shorter than double the width. Solve for x, the width of the tabletop.

59 Emma and Beth do some fundraising for their tennis team and together they raised $\$664$. Emma raised $\$292$ more than Beth. Solve for p, the amount of money raised by Beth.

60 Yvonne is the youngest child in her family. She has three older brothers of ages 23, 20 and 16, and a sister who is eight years older than Yvonne is. The average age of the five children is 19 years. If Yvonne is k years old, find the value of k.

61 Stormie tries to guess how many people are at a concert, but she guesses 800 too many. Pierce guesses 300 too few. The average of their guesses is 4250.

- **a** Describe a method to construct the equation to solve for the exact number of people at the concert.
- **b** Solve for the exact number of people at the concert.

62 Athena and Emilio want to go karting. It cos[redacted]cents per lap of the course.

a Write an equation to solve for the number of laps Athena and Emilio can afford if they have $\$12$.

b Solve for the number of laps they can afford.

c How does that equation change if they have to put a deposit of $\$1$ on each cart used?

63 Bassam is changing km/h into mi/h for a project on peregrine falcons. He knows that the top speed of the peregrine falcon is 389 km/h. He knows that there is 1.6 km to 1 mile. To convert 389 km/h to mi/h, Bassam multiplies 389 by $\dfrac{1.6 \text{ km}}{1 \text{ mi}}$.
What error does Bassam make?

64 Determine which of the following are true or false. Explain your thinking.

a A known solution can be checked by substituting it into the equation.

b No two equations have the same solution.

c Any equation with variables on both sides must have multiple solutions.

d Two equations will always have the same solution if one is a multiple of the other.

65 For each of the following relations:

i Write an equation for the relation, using n to represent the unknown number.

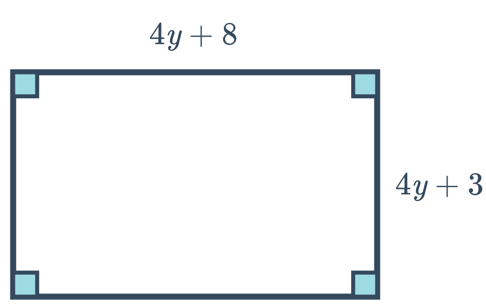
ii Determine the number of solutions to the equation.

a Three more than three times a number is equal to six more than four times the number.

b Six more than two times a number is equal to five more than two times the number.

c Two more than four times a number is equal to two added to quadruple the number.

66 The following rectangle has a perimeter of $126 + 3y$ centimetres:
Find the value of y.



67 A Payroll Officer has been told to distribute a bonus to the employees of a company worth 15% of the company's net income. Since the bonus is an expense to the company, it must be subtracted from the income to determine the net income. If the company has an income of $\$120\,000$ before the bonus, then the Payroll Officer must solve:

$$B = 0.15\left(120\,000 - B\right)$$

for B. Find the bonus, B.

68 Right now, Skye's father is 36 years older than Skye. 4 years ago, her father was 4 times as old as her. Solve for y, Skye's current age.

69 Xanthe, a tennis player, has won 69 out of 96 matches in her career. Find x, the number of matches she must win in a row to raise her win percentage to 75%.

70 Three consecutive integers are such that the sum of the first and twice the second is 15 more than twice the third.

a. Let x be the smallest of the integers. Find x.

b. State the three consecutive integers.

c. Are there any other sets of three consecutive integers that could fulfill the requirements? Explain how you know.